DataExpert.io Community Academy — Week 2 Fact Data Modeling Grader Feedback

Student

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Final Grade

B

Overall Assessment

Thanks for the thorough submission. You completed all eight prompts with clear, well-linted SQL and thoughtful comments explaining grain, behavior, and edge cases. Overall, your modeling choices are sound and consistent, and your incremental patterns are correct.

Global Strengths

• Clear headers and rationale in every file; consistent casing and formatting.

• Correct grains defined and enforced with primary keys.

• Incremental logic covers carry-forward, new entities, and no-activity days.

• Good use of array operations and `GROUP BY` to prevent double-appending of the same day.

• Sensible pre-deduplication of devices to avoid join fan-out.

Global Opportunities

• **Table naming alignment:** Prompt references `nba_game_details` and `web_events`; submission uses `game_details` and `events`. Verify against the actual bootcamp schema.

• Parameterize `current_date` / `snapshot_date` for reusable, idempotent jobs.

• Consider renaming `date` to `snapshot_date` in cumulative tables.

• Consider `ON CONFLICT DO UPDATE` for cumulative inserts where reruns should be safe.

Query 1 — De-duplication

Strengths

• Correct partitioning by `(game_id, team_id, player_id)`.

• Clean expected-column selection, including quoted `"TO"`.

• Readable `ROW_NUMBER()` approach.

Improvements

• Verify `game_details` versus required `nba_game_details`.

• Add a deterministic tie-breaker when duplicate rows differ. Ordering only by partition keys can select an arbitrary duplicate.

• PostgreSQL `DISTINCT ON` could also simplify the pattern.

Query 2 — user_devices_cumulated DDL

Strengths

• Correct grain: user + browser + snapshot date.

• `DATE[]` is an appropriate activity-date representation.

• Primary key enforces the intended grain.

Improvements

• Prefer `snapshot_date` over `date`.

• Optionally prevent unexpected NULL activity arrays with an appropriate constraint/default.

Query 3 — user_devices_cumulated Incremental Population

Strengths

• Deduplicates `devices`, preventing join fan-out.

• `FULL OUTER JOIN` handles carry-forward and new combinations.

• Today is grouped to one `(user, browser)` row, preventing duplicate date appends.

Improvements

• Verify `events` versus `web_events`.

• Consider `ON CONFLICT` upsert behavior for idempotent reruns.

Query 4 — User Devices Activity Integer Datelist

Strengths

• Correct conceptual 32-day window.

• Correct bit direction, with current date at the highest-order bit.

• Avoiding array `UNNEST` is acceptable.

Important Functional Fix 1 — generate_series alias

The grader identified an aliasing problem. Prefer an explicit column alias:

```
CROSS JOIN LATERAL generate_series(
DATE '2023-01-31' - INTERVAL '31 day',
```

```
DATE '2023-01-31',
INTERVAL '1 day'
) AS gs(valid_date)
```

Then use `gs.valid_date`.

Important Functional Fix 2 — Integer type

The requirement calls for a base-2 **integer representation**. The submitted `BIT(32)` result should instead be a `BIGINT`, for example:

```
SUM(
CASE
WHEN is_active
THEN (1::BIGINT << (31 - days_since))
ELSE 0::BIGINT
END
) AS datelist_int
```

Additional improvement

Parameterize the snapshot date so the 32-day calculation is reusable.

Query 5 — hosts_cumulated DDL

Strengths

• Correct grain and primary key.

• `DATE[]` fits the requirement.

Improvement

Prefer `snapshot_date` to the generic `date` column name.

Query 6 — hosts_cumulated Incremental

Strengths

• Correct carry-forward and new-host handling.

• Correct current-date append behavior.

• `FULL OUTER JOIN` covers expected states.

Improvements

• Verify `events` versus `web_events`.

• Consider `ON CONFLICT` handling for idempotent reruns.

Query 7 — host_activity_reduced DDL

Strengths

• Correct `(month, host)` grain and primary key.

• `hit_array` and `unique_visitors` use required `BIGINT[]`.

• Array semantics are clearly documented.

Query 8 — host_activity_reduced Incremental

Strengths

• Excellent day alignment and zero-filling.

• `ARRAY_FILL` correctly creates leading zeros for hosts first appearing after day one.

• `FULL OUTER JOIN` preserves inactive-but-known hosts with zero appends.

• `ON CONFLICT` makes this fact-table process idempotent.

• Date parameterization is already used here.

Improvement

Verify `events` versus `web_events`.

Schema and Environment Confirmation

The grader requested confirmation of:

• `nba_game_details` versus `game_details`

• `web_events` versus `events`

• DDL/columns for the game-details, web-events, and devices sources

• Target warehouse/SQL dialect

• Intended snapshot-date and rerun policy

Verdict

All eight prompts were implemented with strong clarity and largely correct logic. The main functional issue is Query 4's aliasing/type implementation, plus possible source-table naming misalignment.

Final Grade: B

Remediation Priorities

1. Correct Query 4 `generate_series` aliasing.

2. Change Query 4 output from `BIT(32)` to `BIGINT`.

3. Verify/align `nba_game_details` versus `game_details`.
4. Verify/align `web_events` versus `events`.
5. Parameterize snapshot dates consistently.
6. Add explicit rerun/idempotency handling where appropriate.
7. Re-run SQL validation and preserve evidence.
8. Build and integrity-check a fresh eight-query resubmission ZIP.